Immunosenescence and cancer: mechanisms and therapeutic targets

ABSTRACT

Immunosenescence contributes to tumor progression. This review summarizes current knowledge on immune aging and cancer immunotherapy.

Keywords: Immunosenescence; Tumor progression; Aging

Concept, process, and hallmarks of immunosenescence

Immunosenescence is a multifactorial process. Its hallmarks include cell cycle arrest, telomere shortening and altered signaling.

Immunosenescence and cancer

Generally, the risk of malignant tumors increases with age. In addition to the accumulation of genetic mutations, many researchers believe that immunosenescence may also play an important role in the tumoral process.

Table 2 List of the current clinical trials targeting immunosenescence

Drug	Target	Phase	References
Rapamycin	mTOR	2	[41, 42]
TAB08	NCT01990157	1	[55]
Metformin	AMPK	3	[60]

Fig. 1 The process of immunosenescence. The process of immunosenescence can alter the immune response, thus leading to the occurrence of various diseases, such as tumors and infections.

DeSantis et al. [49] found a higher risk of tumorigenesis in the elderly group. By analyzing the age distributions of 100 different tumors, Palmer et al. [24] concluded that the immune system may play an important role in tumor development.

Conclusion

Immunosenescence is a key driver of tumor progression.

Table 3 Supplementary summary
Note: extra row

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References

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